

Cloud Networking – Assignment 5

1. Pick any one app you use daily (like Zomato, BookMyShow, or Spotify) and write a short description of a new feature you would want to build for it that would require cloud hosting and scaling.

Ans: App: Zomato

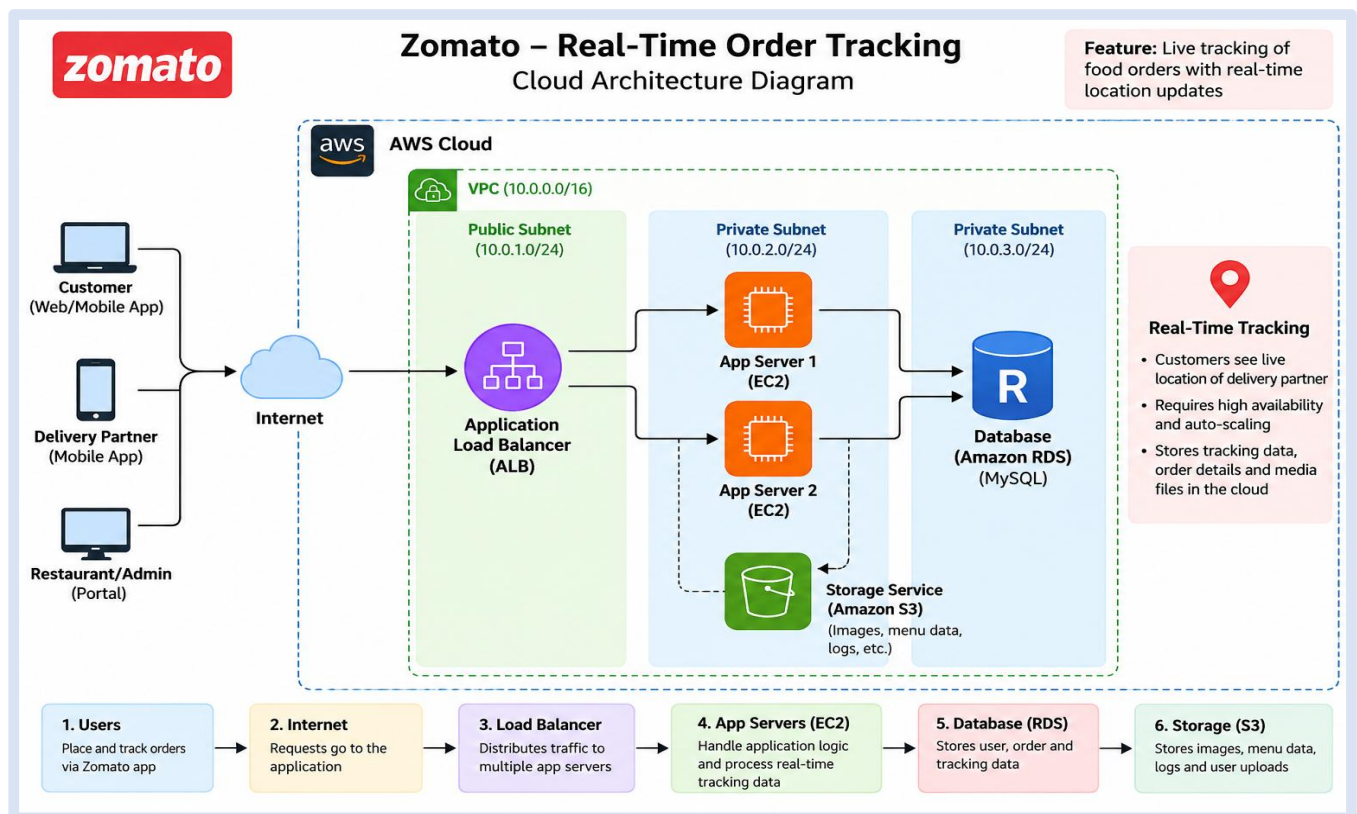
Feature: Real-Time Order Tracking

I would like to add a real-time order tracking feature to Zomato. Users can see the delivery partner's live location and the estimated arrival time on a map.

This feature requires cloud hosting and scaling because many users may track their orders at the same time, especially during peak hours. Cloud scaling can automatically add more resources when the number of users increases and reduce them when the traffic decreases.

2. Draw a basic cloud architecture diagram (hand-drawn or using a tool like draw.io) for your chosen feature, including at least: a load balancer, two app servers, a database, and a storage service (like AWS S3 or Azure Blob). Label each component clearly.

Ans:



3. List three security measures you would add to your architecture to protect user data and prevent attacks. For each, explain in 1-2 lines how it helps (e.g., HTTPS, firewalls, authentication methods).

Ans: Security Measures for the Architecture

1. HTTPS / TLS

Use HTTPS to encrypt communication between users and the application. It protects sensitive data such as login details and order information from being intercepted.

2. Security Groups and Network ACLs

Use Security Groups and Network ACLs to allow only required network traffic. This helps block unauthorized access to the application servers and database.

3. Authentication and Authorization

Use secure user authentication with passwords and tokens, and allow users to access only the data and features they are authorized to use. This helps prevent unauthorized access to user accounts and data.

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4. Write a short document (200-300 words) explaining how your architecture would handle a sudden spike in users during a big event (like IPL finals for BookMyShow or a major album release on Spotify). Describe which parts would scale and how.

Ans:

Handling a Sudden Spike in Users

The proposed **Zomato real-time order tracking architecture** is designed to handle a sudden increase in users during peak hours or a special event. When many users open the app and track their orders at the same time, the **Application Load Balancer (ALB)** receives the incoming requests and distributes them across multiple app servers. This prevents one server from becoming overloaded.

The **EC2 app servers** are the main components that scale. An **Auto Scaling Group** can monitor CPU utilization and traffic. When the workload increases, it automatically launches additional EC2 instances. The load balancer then distributes new requests among the available servers. When the traffic decreases, unnecessary instances can be removed to reduce resource usage and cost.

The **Amazon RDS database** stores important information such as user details, order details, and real-time tracking data. It can be configured with appropriate database scaling and high availability so that it can handle increased requests.

Amazon S3 is used to store images, menu data, and other files. Since S3 is highly scalable, it can handle a large number of requests without manually adding servers.

Security Groups and HTTPS protect the application while handling the increased traffic. Overall, the combination of **Load Balancer, Auto Scaling, EC2, RDS, and S3** allows the application to remain available and responsive even when the number of users suddenly increases.

- 5. Present your architecture design to a friend or family member, ask them for one piece of feedback or a question, and write down what they said along with your response. Hint: Focus on explaining your choices simply, as if you were talking to someone new to cloud computing.**

Ans:

I presented my Zomato Real-Time Order Tracking architecture to a friend and explained how the Load Balancer, EC2 servers, RDS database, and S3 storage work together.

Feedback/Question:

“My question is, what happens if many users track their orders at the same time?”

My Response:

The Load Balancer distributes user requests between multiple EC2 servers. If the traffic becomes very high, Auto Scaling launches additional servers to handle the extra load. This keeps the application responsive and available.